

# Final report: Smart heating control system



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As part of the 'Smart Home Practical' module, an innovative and user-friendly solution for heating control in work environments was developed. The aim was to increase the energy efficiency of the heating by intelligently controlling the radiators and at the same time improve individual comfort for employees and users of the control system.

The heating control is based on a shared calendar of a Nextcloud instance and enables personalised room temperature control that intelligently takes into account the presence of employees. This technology allows team members to set their preferred room temperature in advance via a web app and the heating control system automates the temperature in the building according to the attendance times in the calendar. Particular attention was paid to the digital accessibility of the web app, which ensures that as many users as possible have barrier-free access to the application.

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## 1. Project objectives

The primary objectives of our project were:

1. **Increase energy efficiency:** The smart heating control system is designed to reduce energy consumption in office and work environments by adapting heating usage to employee attendance times. This should ensure better efficiency, as the automatic adjustment of the room temperature avoids unnecessary heating
2. **Improving individual comfort:** Each user can set their preferred room temperature on the web, combined with calendar synchronisation of attendance times, the heating control system ensures that the workplace is already at the desired temperature when they arrive and that the heating is switched off when they leave. No heating needs to be regulated manually.
3. **Promoting digital accessibility:** The web interface of the heating control system has been developed to be accessible to all users, regardless of their individual needs.
4. **Integration into existing systems:** By using Nextcloud as the basis for the shared calendar and utilising a Raspberry Pi in combination with the Fritz!Box for control, a solution has been created that integrates seamlessly with existing systems and can address multiple heaters.

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## 2. Functionality

The heating control system maps the following functions:

### 2.1 Attendance-based temperature control

Employees enter their attendance times in a shared Nextcloud calendar. Based on the entries, the room temperature in the controlled building is adjusted to the individual needs of the users. If several employees are present in a room, an average temperature is determined and set from the employees' set preferred temperatures. If no user is present, the standard temperature from the system settings is set.

### 2.2 User-based setting options

Every employee in the building can log in to the heating control system independently and define a preferred temperature for their room. They have several options for setting the interface according to their wishes and accessibility requirements. Each user can choose between day and night mode, sepia mode, font size and line spacing options. An English version is also available.

### 2.3 Visual control dashboard

On the dashboard, all users can view the setup of the thermostats and the rooms of the system in an easily accessible and clear manner. The target temperature and the current temperature of the radiators can be clearly viewed. It is also possible for anyone to set the heaters manually.

### 2.4 Automatic calendar event assignment

The events in the employees' shared calendar are analysed by the heating control system to ensure presence-based temperature control. To do this, each user can define a calendar string that assigns events to them in the calendar. In this way, the system remains variable for future changes to the maintenance of the calendar.

### 2.5 Intelligent temperature control

Heating control is provided by a Raspberry Pi, which automatically regulates the temperature of several rooms in the building in contact with the Fritz! In order to take all factors into account in the control, a special algorithm has been developed that regulates the temperature based on the presence, settings and room of all users.

### 2.6 System management

New radiator thermostats can be added to and removed from the system by the administrator via a web interface. Each administrator can manage the default temperature of the system, the preheating time and the night-time switch-off.

### 2.6.1 Configurable preheating

In the system settings, an administrator can define the preheating time in minutes per degree. This allows the system to preheat the building before an employee arrives in the morning.

### 2.6.2 Night-time switch-off

A switch-off time can be set on an hourly basis for the night and times of day when no employees are in the building. The standard temperature is automatically set at this time.

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## 3. Technologies used

The following technologies were used to implement the project:

**Astro.js:** The front-end JavaScript framework Astro.js makes it possible to create a multi-platform, component-based and SEO-friendly website while remaining easy to use. Svelte is used for dynamic components.

**Express.js:** As a backend application, Express.js is used as a Node.js runtime in order to be able to build on a simple, developer-friendly solution with little boilerplate.

**Nextcloud:** We use the open source solution Nextcloud to manage attendance times. The calendar function is accessible via an HTTP interface.

**MongoDB:** To store our data and settings, we use MongoDB, a NoSQL database that is characterised by high performance and simplicity.

**Docker & Docker Compose:** To ensure easy installation, maintenance and deployment, the system is provided as a Docker Compose stack.

**Raspberry Pi:** The Raspberry PI serves as the central server of the system, as it enables a local connection to the Fritz!Box and provides sufficient computing power for the application.

**Fritz!Box:** We use the Fritz!Box to integrate the existing FRITZ!DECT 301/302 smart home devices; an HTTP interface is available in the local network, the Fritz-AHA-API.

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## 4. Outlook

In the future, smart heating control could be further developed with additional functions. These include:

- **Extended automation:** The integration of weather data and other sensor data should make heating control even more efficient and flexible. For example, the

outdoor temperature could be incorporated into the control system to dynamically adjust the heating output.

- **Smartphone-App:** The development of a mobile web app is possible with the Astro front-end framework. This allows users to manage their settings on the move. The mobile solution can currently be used, but is not optimised.
  - **Integration of other smart home components:** In addition to heating control, lighting and air conditioning systems could also be integrated into the system to enable holistic control of the office infrastructure.
  - **Enhanced accessibility:** Accessibility could be further improved and offer additional support for various disabilities, such as voice control for visually impaired users.
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## 5. Conclusion

The smart home heating control system shows how modern technology can be used to improve comfort and energy efficiency in working environments. By integrating it into existing systems and taking into account the needs of all users, we have developed a solution that is both functional and inclusive.